



Semester: July 2024 –November 2024		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 01	Class: TY	Semester:V (SVU 2020)
Programme: Computer Engineering		
Institute/School/Department: K J Somaiya School of Engineering		Name of the department:COMP
Course Code: 116U01C503	Name of the Course: Operating Systems	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory		
3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Explain the system booting process with a neat diagram.	5
ii)	Define the following parameters with respect to choice of an appropriate scheduling algorithm – Throughput, Efficiency, Fairness, Turnaround time, Response time.	5
iii)	What is demand paging? State any two advantages and disadvantages of demand paging scheme of memory management.	5
iv)	Discuss evolution of OS.	5
v)	Explain the sequential and indexed-sequential file organization methods.	5
vi)	Discuss binary and counting semaphores. Write pseudocode for wait and signal semaphore primitives.	5

Que. No.	Question	Max. Marks										
Q2 A	Solve the following	10										
i)	What is multithreading? Can user level threads support multithreading? Justify your answer.	5										
ii)	Explain how the PCB is used during context switching between processes. Also describe why PCBs are critical to process management in a multitasking operating system.	5										
	OR											
Q2 A	Consider the set of process CPU burst time (in milliseconds) shown below. <table border="1"><thead><tr><th>Process</th><th>Burst Time</th></tr></thead><tbody><tr><td>P0</td><td>26</td></tr><tr><td>P1</td><td>3</td></tr><tr><td>P2</td><td>7</td></tr><tr><td>P3</td><td>2</td></tr></tbody></table> Illustrate the scheduling of processes using First Come First Served (FCFS) with the help of Gantt charts under the following conditions: A. All processes arrive at the same time. B. Processes arrive according to the shortest burst time. Analyse the issues in above scenario. Comment on average waiting time.	Process	Burst Time	P0	26	P1	3	P2	7	P3	2	10
Process	Burst Time											
P0	26											
P1	3											
P2	7											
P3	2											

Q 2 B	Solve any One	10
i)	Consider the sleeping barber problem- Imagine a barbershop with a single barber and a row of waiting chairs for customers. The barber spends most of his time cutting hair, but when there are no customers, he takes a nap in the barber chair. When a customer arrives at the barbershop and finds the barber sleeping, they wake him up and get a haircut. If other customers arrive while the barber is busy, they either wait in the chairs if there are empty seats or leave if all the chairs are occupied. The key challenge in this scenario is to ensure that customers are served in a fair and orderly manner, without overcrowding the shop or having customers wait indefinitely. A. Discuss issues with respect to process synchronization and concurrency. B. If a solution is to be designed for this problem using semaphore and/or mutex, List items/objects for which semaphores could be used. <u>State</u> types of suggested semaphores. C. Justify your semaphore choices.	10
ii)	For an instance of readers-writer' s problem involving many readers and many writers- A. Discuss and justify issues with respect to process synchronization and concurrency. B. If a solution is to be designed for this problem using monitors, which conditional variables would be more significant? Discuss your approach.	10

Que. No.	Question	Max. Marks																																																																																											
Q3	Solve any Two	20																																																																																											
i)	Consider the given instance of Banker's algorithm. <table><tr><th></th><th colspan="4">Allocation</th><th colspan="4">Request</th><th colspan="4">Available</th></tr><tr><th></th><th>R₁</th><th>R₂</th><th>R₃</th><th>R₄</th><th>R₁</th><th>R₂</th><th>R₃</th><th>R₄</th><th>R₁</th><th>R₂</th><th>R₃</th><th>R₄</th></tr><tr><td>P₁</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>2</td><td>0</td><td>0</td><td>0</td></tr><tr><td>P₂</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td></td><td></td><td></td><td></td></tr><tr><td>P₃</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td></td><td></td><td></td><td></td></tr><tr><td>P₄</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td></td><td></td><td></td><td></td></tr><tr><td>P₅</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td></td><td></td><td></td><td></td></tr></table> A. Apply the Banker's algorithm to figure out if the system is in safe state or unsafe state. B. If the system is in safe state, give sequence of process execution.		Allocation				Request				Available					R ₁	R ₂	R ₃	R ₄	R ₁	R ₂	R ₃	R ₄	R ₁	R ₂	R ₃	R ₄	P ₁	1	0	0	0	0	1	0	0	2	0	0	0	P ₂	0	1	0	0	0	0	1	0					P ₃	0	0	1	0	0	0	0	1					P ₄	0	1	0	1	1	0	0	0					P ₅	0	0	0	1	0	0	0	0					10
	Allocation				Request				Available																																																																																				
	R ₁	R ₂	R ₃	R ₄	R ₁	R ₂	R ₃	R ₄	R ₁	R ₂	R ₃	R ₄																																																																																	
P ₁	1	0	0	0	0	1	0	0	2	0	0	0																																																																																	
P ₂	0	1	0	0	0	0	1	0																																																																																					
P ₃	0	0	1	0	0	0	0	1																																																																																					
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P ₅	0	0	0	1	0	0	0	0																																																																																					
ii)	Explain the concepts of main memory, physical memory, logical memory, and virtual memory in detail. Discuss the relationship between these different types of memory and how they interact within a modern operating system.	10																																																																																											
iii)	Discuss various hardware approaches for concurrency and synchronization. Should they be a recommended practice? Justify your answer.	10																																																																																											

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Consider a disk with tracks numbered from 0 to 199. The I/O requests are received for tracks- 98, 183, 41, 122, 14, 124, 65, and 67. The read-write head starts at track 53. A. Calculate the total head movements and average seek time using the disk scheduling algorithms: a. First-Come, First-Serve (FCFS) b. SCAN B. Illustrate the head movement with a suitable graph.	10
ii)	A. What are the file inodes? Discuss contents of Unix file inode. B. Assuming that a disk block structure can hold 10 direct memory addresses for a file, explain with a neat diagram how the file management module can support files of very large sizes.	10
iii)	A. Discuss various file Directory structures in terms of their characteristics. B. For each structure, identify the advantages and disadvantages in terms of usability, access efficiency, and data organization	10

Que. No.	Question	Max. Marks
Q5	Solve any Four	20
i)	Discuss device drivers as interface between OS and hardware.	5
ii)	What is thrashing? How does it affect performance?	5
iii)	Discuss the necessary and sufficient condition for a deadlock to occur in system?	5
iv)	Discuss the advantages and disadvantages of FIFO and LRU page replacement policies.	5
v)	Explain circular I/O buffering technique. State applications of the same.	5
vi)	Compare and contrast best fit and worst fit memory allocation methods. Support your answer with a suitable example.	5



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29.05.2024 (E)

Maximum Marks: 100		Semester: MAY-2024	Examination: ESE Examination - KT	Duration: 3 Hrs.
Programme code: 01		Programme: B.Tech in Computer Engg.	Class: TY	Semester: V (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering			Name of the department: Computer Engineering	
Course Code: 116U01C503		Name of the Course: Operating System		
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary				

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	What is the critical section? What are the minimum requirements that should be satisfied by a solution to critical section problem?	5
ii)	Explain Free Space Management.	5
iii)	Give Peterson solution to the Critical section Problem. Give its limitation.	5
iv)	Differentiate between time sharing and real time operating system.	5
v)	Define process. Give the structure of process in Memory.	5
vi)	What is Medium Term scheduler. Explain its role in the Computer system.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Explain different Multithreading Models.	5
ii)	What are the various file accessing methods? Explain them.	5
	OR	
Q2 A	Explain the problem of Busy Waiting. Give the solution to this problem in Semaphore.	10
Q 2 B	Solve any One	10
i)	Explain the different schemes for defining the logical structure of a directory?	10
ii)	Explain about Linux's Completely-Fair scheduler in detail.	10

Que. No.	Question	Max. Marks																					
Q3	Solve any Two	20																					
i)	Assume the following workload in a system:	10																					
	<table border="1"> <thead> <tr> <th>Process</th><th>Arrival Time</th><th>Burst Time</th></tr> </thead> <tbody> <tr> <td>P1</td><td>5</td><td>5</td></tr> <tr> <td>P2</td><td>4</td><td>6</td></tr> <tr> <td>P3</td><td>3</td><td>7</td></tr> <tr> <td>P4</td><td>1</td><td>9</td></tr> <tr> <td>P5</td><td>2</td><td>2</td></tr> <tr> <td>P6</td><td>6</td><td>3</td></tr> </tbody> </table>	Process	Arrival Time	Burst Time	P1	5	5	P2	4	6	P3	3	7	P4	1	9	P5	2	2	P6	6	3	
Process	Arrival Time	Burst Time																					
P1	5	5																					
P2	4	6																					
P3	3	7																					
P4	1	9																					
P5	2	2																					
P6	6	3																					

	Draw a Gantt chart illustrating the execution of these jobs using Round robin scheduling algorithm and also Calculate the average waiting time and average turnaround time.	
ii)	List various structures of Page Table and Explain them in detail with supporting diagrams.	10
iii)	With respect to multiple Processor Scheduling, Explain: - Symmetric and Asymmetric processing - Process affinity - Load balancing	10

Que. No.	Question	Max. Marks																												
Q4	Solve any Two	20																												
i)	A disk drive has 200 cylinders, numbered 0 to 199. The drive is currently serving a request at cylinder 53. The queue of pending requests, in order is 98, 183, 37, 122, 14, 124, 65, 67. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk-scheduling algorithms? i) FCFS ii) SSTF iii) SCAN iv) LOOK v) C-SCAN	10																												
ii)	Consider the following snapshot of a system: <table><tr><td></td><td><u>Allocation</u></td><td><u>Max</u></td><td><u>Available</u></td></tr><tr><td></td><td><u>A B C D</u></td><td><u>A B C D</u></td><td><u>A B C D</u></td></tr><tr><td>P₀</td><td>0 0 1 2</td><td>0 0 1 2</td><td>1 5 2 0</td></tr><tr><td>P₁</td><td>1 0 0 0</td><td>1 7 5 0</td><td></td></tr><tr><td>P₂</td><td>1 3 5 4</td><td>2 3 5 6</td><td></td></tr><tr><td>P₃</td><td>0 6 3 2</td><td>0 6 5 2</td><td></td></tr><tr><td>P₄</td><td>0 0 1 4</td><td>0 6 5 6</td><td></td></tr></table> Answer the following questions using the banker's algorithm: a. What is the content of the matrix <i>Need</i>? b. Is the system in a safe state? c. If a request from process P₁ arrives for (0,4,2,0), can the request be granted immediately?		<u>Allocation</u>	<u>Max</u>	<u>Available</u>		<u>A B C D</u>	<u>A B C D</u>	<u>A B C D</u>	P ₀	0 0 1 2	0 0 1 2	1 5 2 0	P ₁	1 0 0 0	1 7 5 0		P ₂	1 3 5 4	2 3 5 6		P ₃	0 6 3 2	0 6 5 2		P ₄	0 0 1 4	0 6 5 6		10
	<u>Allocation</u>	<u>Max</u>	<u>Available</u>																											
	<u>A B C D</u>	<u>A B C D</u>	<u>A B C D</u>																											
P ₀	0 0 1 2	0 0 1 2	1 5 2 0																											
P ₁	1 0 0 0	1 7 5 0																												
P ₂	1 3 5 4	2 3 5 6																												
P ₃	0 6 3 2	0 6 5 2																												
P ₄	0 0 1 4	0 6 5 6																												
iii)	Define the hardware instruction Test And Set() and Swap() also give the mutual exclusion implementation with both.	10																												

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	Discuss Resource allocation graph with respect to deadlock.	5
ii)	"Operating system is resource manager"-Justify this statement with suitable functionality of OS.	5
iii)	Give the differences between logical and physical addresses.	5
iv)	Explain system Boot Process.	5
v)	Differentiate between multilevel queues and multilevel feedback queues.	5
vi)	What is a System call? Explain in detail the system call sequence to copy the contents of one file to another file.	5



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13-6-2023 (E)

Semester: January 2023 –May 2023		
Maximum Marks: 100	Examination: ESE Examination(KT)	Duration:3 Hrs.
Programme code: 01	Class: TY	Semester: V(SVU 2020)
Programme: B. Tech in Computer Engineering		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: COMP
Course Code: 116U01C503	Name of the Course: Operating System	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	With respect to System Software, Describe assembler and loaders.	5
ii)	Differentiate between Compiler and Interpreter	5
iii)	Illustrate the Process State Transition Diagram with suitable diagram.	5
iv)	Explain the concept of Resource pre-emption for Recovery from Deadlock	5
v)	With respect to Linux Operating System, Define Inodes with suitable diagram.	5
vi)	With respect to Memory Management, Describe Internal and External Fragmentation.	5

Que. No.	Question	Max. Marks
Q2	Solve the following	10
A		
i)	Describe the System Boot Process.	5
ii)	Define shell. Further Comment on the different types of shells.	5
	OR	
Q2	Illustrate the following types of Operating System Structures with the help of suitable examples:-	10
A	i. Traditional UNIX System Structure ii. Layered Approach iii. Microkernel System Structure	
Q 2	Solve any One	10
B		
i)	Compare and Contrast between Multilevel Queue and Multilevel Feedback Queue Scheduling Algorithms with the help of suitable diagrams.	10
ii)	Differentiate between User level Threads and Kernel Level Threads. Further Describe the various Multithreading Models.	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	Consider the methods used by processes P1 and P2 for accessing their critical sections whenever needed, as given below. The initial values of shared Boolean variables S1 and S2 are randomly assigned.	10
	Method Used by P1 while (S1 == S2) ;	

	Critical Section S1 = S2; Method Used by P2 while (S1 != S2) ; Critical Section S2 = not (S1); Analyse the methods and State whether Mutual Exclusion and Progress Requirement are being satisfied or not. Justify your answer.	
ii)	Discuss the Semaphore solution for Dining Philosophers Problem.	10
iii)	With respect to Process Synchronization, Examine the Bounded Buffer Problem and Readers Writer Problem.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Consider a disk queue with requests for I/O to blocks on cylinders in order 43, 33, 127, 87, 17, 99, 20. The head is initially at cylinder number 60, moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. If the following Disk Scheduling algorithms are applied:- a) C-LOOK b) C-SCAN For all the algorithms, Find the order in which the requests will be serviced. Further Calculate the total head movement (in number of cylinders) incurred while servicing these requests.	10
ii)	With respect to File Management, Illustrate the various File Allocation Methods with suitable diagrams.	10
iii)	Discuss the Address Translation Scheme to map Pages into frames with the help of suitable diagrams. Further Explain the concept of Segmentation with an example.	10

Que. No.	Question	Max. Marks
Q5	Solve any four	20
i)	Describe System Calls ,Further list the types of system calls.	5
ii)	Describe Linux Scheduling.	5
iii)	Distinguish between Message Passing and Shared Memory	5
iv)	Explain the concept of Monitors with the help of suitable diagrams.	5
v)	Discuss the various Input/Out Buffering schemes with suitable diagrams for each.	5
vi)	Discuss the structure of Hashed Page Table with the help of suitable diagrams.	5



Semester: July 2023 –October 2023		
Maximum Marks: 100	Examination: ESE Examination	Duration:3 Hrs.
Programme code: 01	Class: TY	Semester: V (SVU 2020)
Programme: B.Tech COMP		
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Computer Engg.	
Course Code: 116U01C503	Name of the Course: Operating System	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory		
3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Define the essential properties of Real Time and Time sharing Operating Systems.	5
ii)	Explain creation and termination operations on Process.	5
iii)	Discuss the various Input/ Output Buffering schemes with suitable diagrams.	5
iv)	Compare the memory organization schemes of contiguous memory allocation, pure segmentation, and pure paging with respect to the following issues: a. External fragmentation b. Internal fragmentation	5
v)	Define process and various states of the process with the help of diagram.	5
vi)	What are cooperating processes? Give atleast three reasons for Interprocess communication.	5
Q2 A	Solve the following	10
i)	Using the below program, identify the values of pid at lines A, B, C, and D. (Assume that the actual pids of the parent and child are 2600 and 2603, respectively.) <pre>#include <sys/types.h> #include <stdio.h> #include <unistd.h> int main() { pid_t pid, pid1; /* fork a child process */ pid = fork(); if (pid < 0) { /* error occurred */ fprintf(stderr, "Fork Failed"); return 1; } else if (pid == 0) { /* child process */ pid1 = getpid(); printf("child: pid = %d", pid); /* A */ printf("child: pid1 = %d", pid1); /* B */ } else { /* parent process */ pid1 = getpid(); printf("parent: pid = %d", pid); /* C */ printf("parent: pid1 = %d", pid1); /* D */ wait(NULL); } return 0; }</pre>	5
ii)	Show that, if the wait () and signal () semaphore operations are not executed	5

	atomically, then mutual exclusion may be violated.	
	OR	
Q2 A	Explain the layered approach to Operating system design? Give the advantages and disadvantages of using the layered approach?	10
Q 2 B	Solve any One	10
i)	With respect to Contiguous memory allocation, Illustrate the working of Best Fit, Worst Fit and First Fit allocation algorithm with suitable examples for each.	10
ii)	Explain Multilevel feedback queue scheduling with example. How it is useful than Multilevel queue scheduling.	10

Que. No.	Question	Max. Marks																					
Q3	Solve any Two	20																					
i)	Assume the following workload in a system: <table border="1" data-bbox="227 862 665 1176"> <thead> <tr> <th>Process</th><th>Arrival Time</th><th>Burst Time</th></tr> </thead> <tbody> <tr> <td>P1</td><td>5</td><td>5</td></tr> <tr> <td>P2</td><td>4</td><td>6</td></tr> <tr> <td>P3</td><td>3</td><td>7</td></tr> <tr> <td>P4</td><td>1</td><td>9</td></tr> <tr> <td>P5</td><td>2</td><td>2</td></tr> <tr> <td>P6</td><td>6</td><td>3</td></tr> </tbody> </table> Draw a Gantt chart illustrating the execution of these jobs using Shortest Remaining Time First (SRTF) algorithm and also Calculate the average waiting time and average turnaround time.	Process	Arrival Time	Burst Time	P1	5	5	P2	4	6	P3	3	7	P4	1	9	P5	2	2	P6	6	3	10
Process	Arrival Time	Burst Time																					
P1	5	5																					
P2	4	6																					
P3	3	7																					
P4	1	9																					
P5	2	2																					
P6	6	3																					
ii)	Paging is a non-contiguous memory allocation scheme. Justify. Explain the role of translation look aside buffer (TLB) in paging.	10																					
iii)	Explain critical section problem. State Readers-Writers Problem and give its solution using Semaphore.	10																					

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Illustrate Inter Process Communication. How to handle it using Shared Memory and Message Passing?	10
ii)	Explain Race conditions with example. Consider a banking system with two functions: deposit (amount) and withdraw (amount). These two functions are passed the amount that is to be deposited or withdrawn from a bank account. Assume a shared bank account exists between a husband and wife and concurrently the husband calls the withdraw() function and the wife calls deposit(). Describe how a race condition is possible and what might be done to prevent the race condition from occurring.	10
iii)	Explain Deadlock Prevention in detail.	10

Que. No.	Question	Max. Marks
Q5	Write short notes on any four	20
i)	Process Control Block	5
ii)	Acyclic graph directories structure	5
iii)	Bit vector for free space management	5
iv)	Peterson solution to the Critical section Problem	5
v)	Recovery from Deadlock	5
vi)	Inverted Page Table	5



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VIDYAVIHAR UNIVERSITY

Semester: August 2022 – December 2022 – Jan-2023		
Maximum Marks: 100	Examination: ESE Examination – Re-Exam Duration: 3 Hrs.	
Programme code: 01	Class: TY	Semester: V (SVU 2020)
Programme: Computer Engineering		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computer
Course Code: 116U01C503	Name of the Course: Operating System	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	What is operating system? List the objectives and functions of operating system	5
ii)	Differentiate between monolithic and structured layer of OS	5
iii)	Draw and explain 7-state diagram of life cycle of a process	5
iv)	Explain semaphores and its different types .	5
v)	Explain hierarchical paging	5
vi)	What is segmentation? State its usage.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Suppose that a disk drive has 5000 cylinders numbered 0 to 4999. The drive is currently serving a request at cylinder 2150 and the previous request was at cylinder 1805. The queue of pending requests in FIFO order is 2069, 1212, 2296, 2800, 544, 1618, 356, 1523, 4965, 3681. Starting from the current head position, what is the total distance(in cylinders) that the disk arm moves to satisfy all pending requests for the following Disk scheduling algorithms (i) SCAN (ii) C-LOOK	5
ii)	Explain any two I/O Buffering techniques	5
OR		
Q2 A	What are the various allocation methods with reference to file systems	10
Q 2 B	Solve any One	10
i)	What are the five major categories of system calls? Explain giving one example from each	10
ii)	Differentiate between Network OS and Distributed OS	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	What are the requirements of mutual exclusion? Give software approaches to achieve mutual exclusion.	10
ii)	What is deadlock? Explain various deadlock prevention techniques.	10
iii)	What is critical section problem? Explain different ways to solve critical section problem	10

Que. No.	Question	Max. Marks																								
Q4	Solve any Two	20																								
i)	Consider the following set of Processes with the length of CPU burst given in ms <table><tr><th>Process</th><th>Burst time</th><th>Priority</th><th>Arrival time</th></tr><tr><td>P1</td><td>27</td><td>5</td><td>0</td></tr><tr><td>P2</td><td>6</td><td>3</td><td>3</td></tr><tr><td>P3</td><td>7</td><td>1</td><td>5</td></tr><tr><td>P4</td><td>10</td><td>2</td><td>10</td></tr><tr><td>P5</td><td>15</td><td>4</td><td>13</td></tr></table> Assume smaller number implies higher priority. (i) Draw the Gantt chart for FCFS, SJF, Priority (Pre-emptive) and Round Robin (Quantum = 2). (ii) Which algorithm results in maximum average waiting time? (iii) Calculate average turnaround time for each of the above algorithms	Process	Burst time	Priority	Arrival time	P1	27	5	0	P2	6	3	3	P3	7	1	5	P4	10	2	10	P5	15	4	13	10
Process	Burst time	Priority	Arrival time																							
P1	27	5	0																							
P2	6	3	3																							
P3	7	1	5																							
P4	10	2	10																							
P5	15	4	13																							
ii)	Explain Multithreaded model of a process..	10																								
iii)	Explain multilevel Queue scheduling	10																								

Que. No.	Question	Max. Marks
Q5	Answer any four	20
i)	Write a short note on File organization.	5
ii)	Explain the role of PCB in Process management. Why are two modes of execution needed	5
iii)	Discuss the semaphore solution for Reader/ Writer Problem.	5
iv)	Write short note on Virtual memory.	5
v)	Explain Demand paging	5
vi)	Discuss the monitor solution for dining philosopher problem	5

27-12-2022 (E)


SOMAIYA
 VIDYAVIHAR UNIVERSITY

Semester: August 2022 – December 2022		Duration: 3hours
Maximum Marks: 100	Examination: ESE Examination	
Programme code: 01	Class: FY/SY/TY/LY	Semester: I/II/III/IV/V/VI/VII/VIII (SVU 2020)
Programme: B Tech Computer Engineering		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: COMP/ETRX/EXTC/IT/MECH
Course Code: 116U01C503	Name of the Course: Operating System	
Instructions: 1) Draw neat diagrams 2) Assume suitable data if necessary		

Questi on No.		Max. Marks																								
Q1 (a)	Differentiate between monolithic and microkernel	05																								
Q1 (b)	Define the term Critical section and Race condition.	05																								
Q1 (c)	What is Process Control Block (PCB)?	05																								
Q1 (d)	Explain the effect of page size on the performance of a process.	05																								
Q2 (a)	Given five memory partitions of 100 KB, 500 KB, 200 KB, 300 KB, and 600 KB (in order), how would each of the first-fit, best-fit, and worst-fit algorithms place processes of 212 KB, 417 KB, 112 KB, and 426 KB (in order)? Which algorithm makes the most efficient use of memory?	10																								
Q2 (b)	What are System calls? List and explain file-handling system calls. OR Explain Process control fork(), exec(), and wait() System calls.	10																								
Q3 (a)	Explain the following in brief:(anyone) 1. Semaphores 2. Scheduling in Linux system	10																								
Q3 (b)	Assume the following processes arrive for execution at the indicated time and the CPU burst time is given in ms. <table><tr><th>Process</th><th>Burst Time</th><th>Arrival Time</th><th>Priority</th></tr><tr><td>P1</td><td>10</td><td>0</td><td>5</td></tr><tr><td>P2</td><td>6</td><td>0</td><td>2</td></tr><tr><td>P3</td><td>7</td><td>1</td><td>4</td></tr><tr><td>P4</td><td>4</td><td>1</td><td>1</td></tr><tr><td>P5</td><td>5</td><td>2</td><td>3</td></tr></table> Find the Average waiting time, and the average turnaround time for FCFS, SJF(Non-preemptive), and Round Robin(Quantum-3ms)	Process	Burst Time	Arrival Time	Priority	P1	10	0	5	P2	6	0	2	P3	7	1	4	P4	4	1	1	P5	5	2	3	10
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Q4 (a)	Explain the difference between Paging and Segmentation. OR Explain the hardware support for Paging.	10																																																																																										
Q4 (b)	On a disk with 1000 cylinders, numbers 0-999, compute the number of tracks the disk arm must move to satisfy all requests in the disk queue. Assume the last request received was at track 345 and the head is moving toward track 0. The queue in the FIFO order contains requests for the following tracks 123, 874, 692, 475, 105, 376. Perform the computation for the following scheduling algorithms: i. FIFO ii. SSTF iii. SCAN	10																																																																																										
Q5 (a)	Consider the given snap of the System <table border="1"><thead><tr><th rowspan="2"></th><th colspan="4">Allocation</th><th colspan="4">Max</th><th colspan="4">Available</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th></tr></thead><tbody><tr><td>P0</td><td>0</td><td>2</td><td>1</td><td>2</td><td>0</td><td>3</td><td>2</td><td>2</td><td>2</td><td>5</td><td>3</td><td>2</td></tr><tr><td>P1</td><td>1</td><td>1</td><td>0</td><td>2</td><td>2</td><td>7</td><td>5</td><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td>P2</td><td>2</td><td>2</td><td>5</td><td>4</td><td>2</td><td>3</td><td>7</td><td>6</td><td></td><td></td><td></td><td></td></tr><tr><td>P3</td><td>0</td><td>3</td><td>1</td><td>2</td><td>1</td><td>6</td><td>4</td><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td>P4</td><td>2</td><td>4</td><td>1</td><td>4</td><td>3</td><td>6</td><td>5</td><td>8</td><td></td><td></td><td></td><td></td></tr></tbody></table> Answer the following questions using Bankers algorithm. 1) What is the content of Matrix need? 2) Is the system in a safe state? 3) If a request from process P1 arrives for (1,3,2,1) can the request be granted immediately?		Allocation				Max				Available				A	B	C	D	A	B	C	D	A	B	C	D	P0	0	2	1	2	0	3	2	2	2	5	3	2	P1	1	1	0	2	2	7	5	2					P2	2	2	5	4	2	3	7	6					P3	0	3	1	2	1	6	4	2					P4	2	4	1	4	3	6	5	8					10
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Q5 (b)	Explain the conditions for deadlock. Suggest techniques to avoid deadlock. OR Explain an algorithm for the producer-consumer problem.	10																																																																																										